

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Data Visualization with Power BI and Tableau

Practical – II(Curiosity Question)

Roll No.: T005	Name: Ritesh Chaubey
Class: TYIT	Batch:1
Date of Assignment: 24/07/2026	Date/Time of Submission:24/07/2026

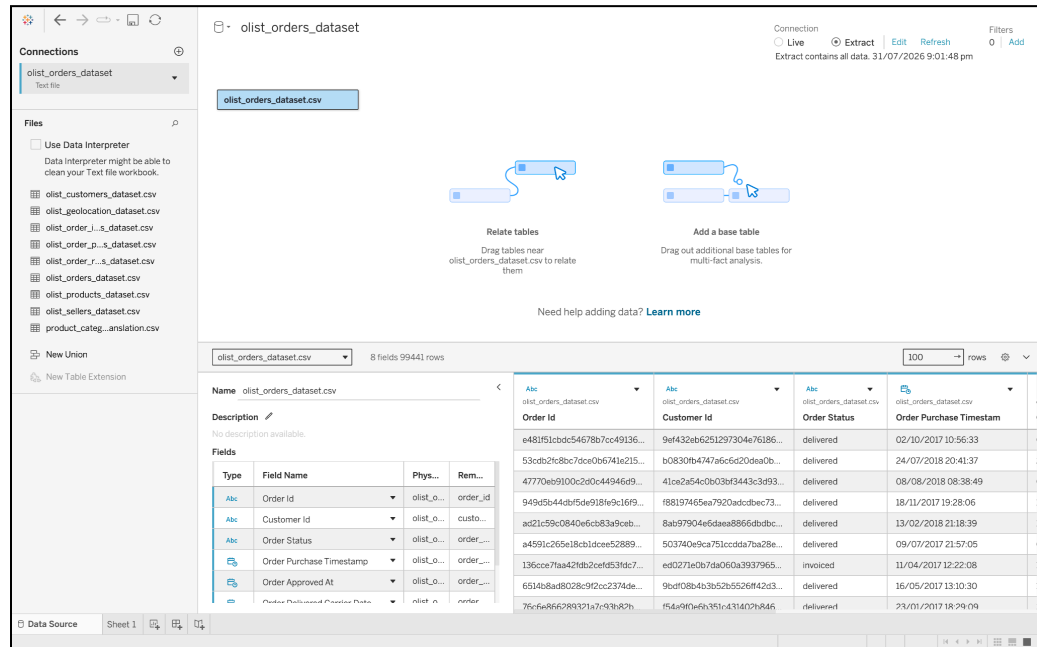
AIM: Working with Data Sources and Data Preparation in Tableau.

- A. Connect to different data sources (Excel, CSV, Database).**
- B. Create and use data extracts vs live connectionsManage metadata (rename, hide, change data types).**
- C. Perform joins (inner, left, right) on multiple tables.**
- D. Apply data blending between two sources.**
- E. Implement different types of filters (dimension, measure, date).**
- F. Optimize performance using extracts and filtering.**

STEPS:

Step 1: Import the Dataset

1. Open **Tableau Desktop**.
2. Under the **Connect** section, click **Text File**.
3. Select **olist_orders_dataset.csv** from your computer and click **Open**.
4. After the dataset is loaded, go to the **Connection** section and select **Extract** instead of **Live**.
5. Click on **Sheet 1**. If prompted, save the extract by entering a suitable file name and click **Save**.
6. Save the workbook by selecting **File → Save As**, choose the **Tableau Packaged Workbook (.twbx)** format, enter a suitable file name, and click **Save**.



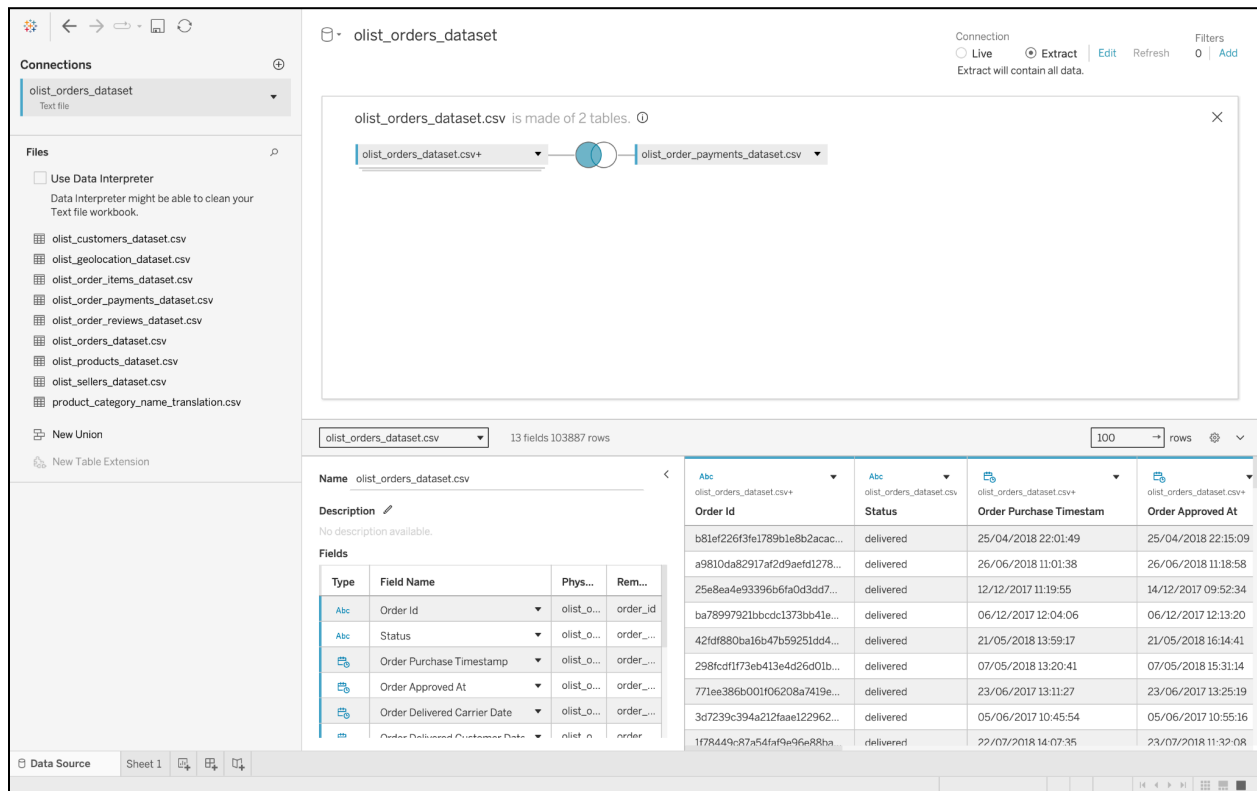
Step 2: Managing Metadata

1. Open **Sheet 1** and locate the required fields in the **Data Pane**.
2. Right-click on **order_status** and select **Rename**. Change the field name to **Status**.
3. Locate the **customer_id** field, right-click on it, and select **Hide** to remove it from the Data Pane.
4. Locate the **order_purchase_timestamp** field and click on its data type icon.
5. If required, change the data type to **Date & Time**. If it is already set as **Date & Time**, no changes are needed.
6. Verify that the metadata changes have been applied successfully.

Step 3: Performing Relational Joins on Multiple Tables

1. Click on the **Data Source** tab at the bottom-left corner to return to the Data Source page.
2. Double-click the **olist_orders_dataset.csv** table to open the **Physical Layer**.
3. From the left pane, drag the **olist_order_payments_dataset.csv** file next to the **olist_orders_dataset.csv** table.
4. Tableau automatically creates a **Join** between the two tables.
5. Click on the **Join** (Venn diagram) icon to open the Join configuration.
6. Select **order_id** as the common field for both tables.
7. Change the join type and observe the results:
8. **Inner Join** – Displays only matching records from both tables.
9. **Left Join** – Displays all records from the Orders table and matching records from the Payments table.
10. Select **Left Join** and keep it as the final join type.

11. Verify that the joined data is displayed correctly in the data preview.



The screenshot displays the Tableau Desktop interface. On the left, the 'Connections' pane shows 'olist_orders_dataset' as a text file. Below it, the 'Files' pane lists various CSV files, including 'olist_orders_dataset.csv'. The main area shows a data preview for 'olist_orders_dataset.csv', which is a union of two tables: 'olist_orders_dataset.csv+' and 'olist_order_payments_dataset.csv'. The preview shows 13 fields and 103887 rows. The data is displayed in a table with columns: Order Id, Status, Order Purchase Timestamp, and Order Approved At. The table contains 10 rows of data.

Order Id	Status	Order Purchase Timestamp	Order Approved At
b81ef226f3fe1789b1e8b2acac...	delivered	25/04/2018 22:01:49	25/04/2018 22:15:09
a9810da82917af2d9aefd1278...	delivered	26/06/2018 11:01:38	26/06/2018 11:18:58
25e8ea4e93396b6fa0d3dd7...	delivered	12/12/2017 11:19:55	14/12/2017 09:52:34
ba78997921bbcdc1373bb41e...	delivered	06/12/2017 12:04:06	06/12/2017 12:13:20
42fd880ba16b47b59251dd4...	delivered	21/05/2018 13:59:17	21/05/2018 16:14:41
298fcd1f73eb413e4d26d01b...	delivered	07/05/2018 13:20:41	07/05/2018 15:31:14
771ee386b001f06208a7419e...	delivered	23/06/2017 13:11:27	23/06/2017 13:25:19
3d7239c394a212faae122962...	delivered	05/06/2017 10:45:54	05/06/2017 10:55:16
1f78449c97af5d1af9e96e88ba...	delivered	22/07/2018 14:07:35	23/07/2018 11:32:08

INSIGHT:

- Relational joins combine data from multiple tables using a common field.
- The joined data provides complete order and payment information.
- It improves data analysis by bringing related records together.

Step 4: Applying Data Blending Between Independent Sources

1. Click on the **New Worksheet** icon at the bottom of the Tableau window to create a new worksheet (**Sheet 2**).
2. In the **Data Pane**, ensure that the **Orders** dataset is selected as the **Primary Data Source**.
3. Drag the **Status** field from the Orders dataset and drop it onto the **Rows** shelf. A **blue check mark** will appear next to the Orders data source, indicating that it has been set as the **Primary Data Source**.
4. From the top menu bar, click **Data** → **Edit Blend Relationships**. If the option is disabled (grayed out), click **Data** → **New Data Source**.
5. In the **Connect** pane, select **Text File** and browse to the location of **olist_customers_dataset.csv**.
6. Select **olist_customers_dataset.csv** and click **Open** to add it as the secondary data source.

7. Return to **Sheet 2** and again click **Data** → **Edit Blend Relationships**.
8. In the **Edit Blend Relationships** dialog box:
9. Set **Primary Data Source** as **olist_orders_dataset**.
10. Set **Secondary Data Source** as **olist_customers_dataset**.
11. Change the relationship mode from **Automatic** to **Custom**.
12. Click the **Add** button to create a custom relationship between the two datasets.
13. Select **customer_id** from the Orders dataset as the primary field.
14. Select **customer_id** from the Customers dataset as the secondary field.
15. Click **OK** to save the relationship and then click **OK** again to close the dialog box.
16. Go back to the **Data Source** page and change the connection mode from **Extract** to **Live**.
17. Return to **Sheet 2**.
18. In the **Data Pane**, right-click on an empty space and select **Show Hidden Fields** so that hidden fields become visible again.
19. Locate the **customer_id** field and click the **chain link** icon beside it. The chain icon should turn **orange**, indicating that the blend relationship is active between the two data sources.
20. Select the **Customers** dataset from the Data Pane.
21. Drag the **customer_state** field from the Customers dataset and drop it onto the **Columns** shelf.
22. Verify that data from both the **Orders** and **Customers** datasets is displayed correctly in the worksheet, confirming that the data blending process has been completed successfully.

Blend Relationships

Blend relationships determine how data from secondary data sources are joined with primary data sources.

Primary data source:

olist_orders_dataset

Secondary data source:

olist_customers_dataset

☒ Automatic

☐ Custom

Add...

Edit...

Remove

Cancel

OK

Add/Edit Field Mapping

Primary data source field:

Search

Customer Id

> Order Approved At

Order Approved At

> Order Delivered Carrier Date

Order Delivered Carrier Date

> Order Delivered Customer Date

Order Delivered Customer Date

> Order Estimated Delivery Date

Order Estimated Delivery Date

Order Id

Order Id (Olist Order Payments Data

> Order Purchase Timestamp

Order Purchase Timestamp

Payment Type

Status

Table Name

Secondary data source field:

Search

Customer City

Customer Id

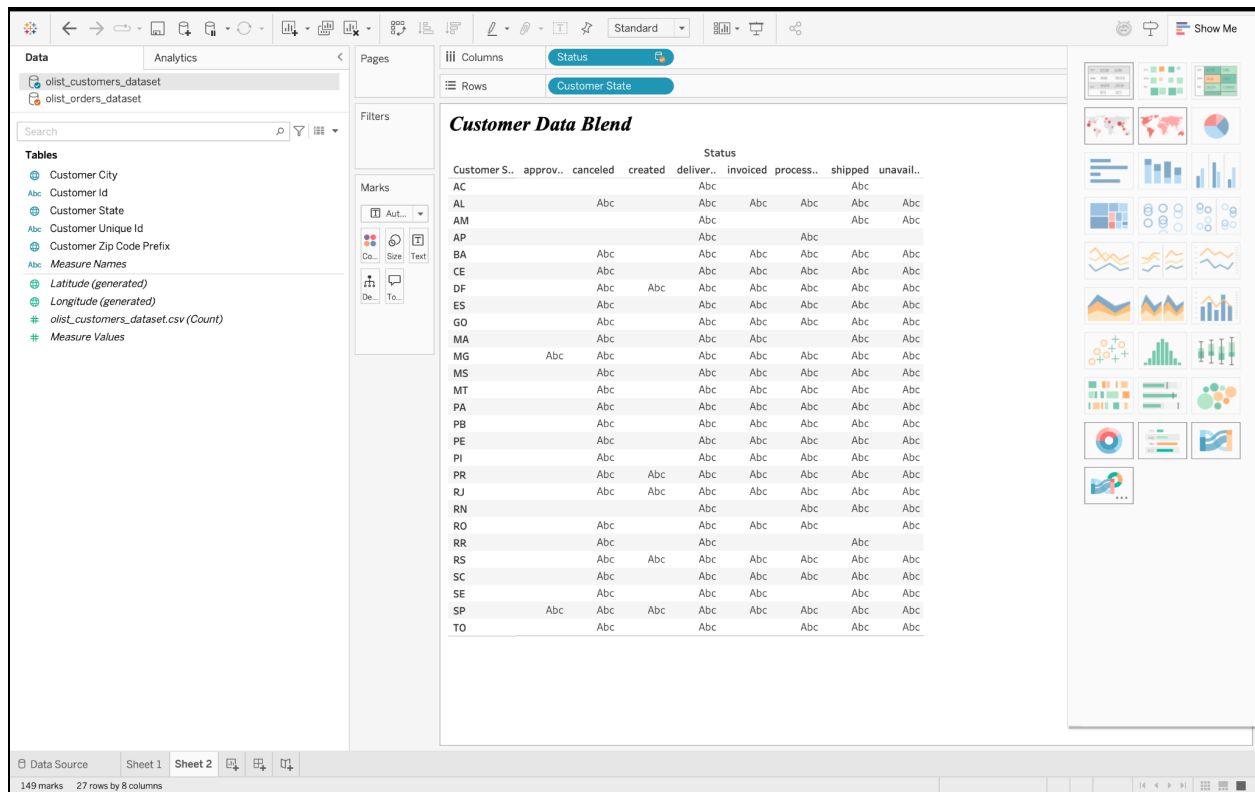
Customer State

Customer Unique Id

Customer Zip Code Prefix

Cancel

OK



INSIGHT:

- Data from the **Orders** and **Customers** datasets has been blended successfully.
- The dashboard displays **Customer State** along with **Order Status**.
- Data blending enables analysis of customer information and order details together.

Step 5: Creating Context Filters.

1. Open a **new worksheet (Sheet 3)** in Tableau.
2. From the **Data Pane**, drag the **Status** field and drop it onto the **Filters** shelf.
3. The **Filter [Status]** dialog box will appear.
4. Under the **General** tab, keep all status values selected (**approved, canceled, created, delivered, invoiced, processing, shipped, and unavailable**) and click **OK**.
5. Right-click the **Status** filter on the **Filters** shelf and select **Show Filter**. The **Status Filter Card** will appear on the right side of the worksheet.
6. Next, drag the **Payment Value** field from the Data Pane to the **Filters** shelf.
7. The **Filter Field [Payment Value]** dialog box will appear. Select **All Values** and click **Next**.
8. In the **Filter [Payment Value]** window, keep the **Range of Values** option selected. Leave the default minimum and maximum values unchanged and click **OK**.
9. Right-click the **Payment Value** filter and select **Show Filter**. A **range slider** for Payment Value will appear on the right side of the worksheet.

10. Now drag the **Order Purchase Timestamp** field from the Data Pane to the **Filters** shelf.
11. The **Filter Field [Order Purchase Timestamp]** dialog box will open. Select **Range of Dates** and click **Next**.
12. In the **Filter [Order Purchase Timestamp]** window, keep the default **Start Date** and **End Date** values, then click **OK**.
13. Right-click the **Order Purchase Timestamp** filter and select **Show Filter**. A **date range filter** will appear on the right side of the worksheet.
14. Verify that all three filters (**Status**, **Payment Value**, and **Order Purchase Timestamp**) are displayed on the worksheet and can be used to interactively filter the data.

Filter [Status]

General Wildcard Condition Top

☒ Select from list ☐ Custom value list ☐ Use all

Enter search text

- ☒ approved
- ☒ canceled
- ☒ created
- ☒ delivered
- ☒ invoiced
- ☒ processing
- ☒ shipped
- ☒ unavailable

All None ☐ Exclude

Summary

Field: [Status]
Selection: Selected 8 of 8 values
Wildcard: All
Condition: None
Limit: None

Reset Apply Cancel OK

[illegible]

Filter Field [Payment Value]

How do you want to filter on [Payment Value]?

- # All values
- # Sum
- # Average
- # Median
- # Count
- # Count (Distinct)
- # Minimum
- # Maximum
- # Standard deviation
- # Standard deviation (Population)
- # Variance
- # Variance (Population)
- # Attribute

Filter [Payment Value]

☒ Range of values ☐ At least ☐ At most ☐ Special

Range of values

0 13,664.08

0 13664.08

Show: ☐ Include Null Values

[illegible]

Filter Field [Order Purchase Timestamp]

How do you want to filter on [Order Purchase Timestamp]?

- Relative Date
- Range of Dates**
- Years
- Quarters
- Months
- Days
- Hours
- Minutes
- Seconds
- Week numbers
- Weekdays
- Month / Year
- Month / Day / Year
- Individual Dates and Times
- Count
- Count (Distinct)
- Minimum
- Maximum
- Attribute

Cancel Next >

Filter [Order Purchase Timestamp]

Relative dates Range of dates Starting date Ending date Special

Range of dates

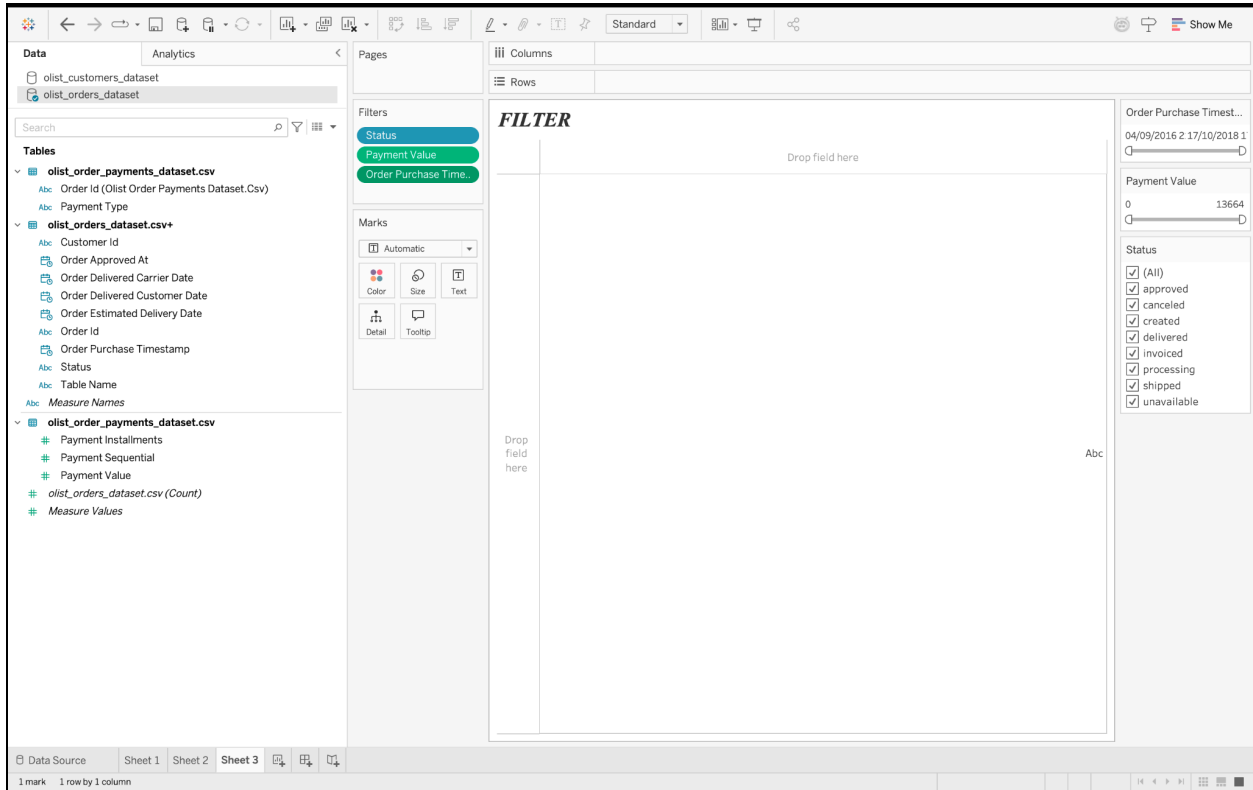
☒ Show Times

4/9/2016, 9:15:19PM 17/10/2018, 5:30:18PM

04/09/2016 21:15:19 17/10/2018 17:30:18

Show: Only Relevant Values ☐ Include Null Values

Reset Apply Cancel OK



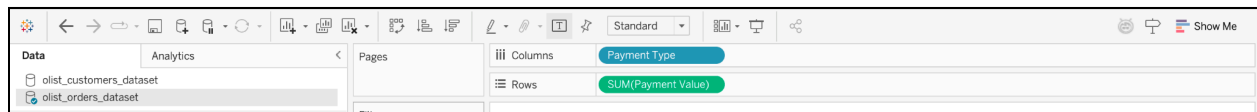
INSIGHTS:

- Filters allow users to view only the required data.
- The **Status** filter displays records based on the selected order status.
- The **Payment Value** filter helps analyze orders within a selected payment range.
- The **Order Purchase Timestamp** filter allows analysis for a specific date range.
- Interactive filters make the dashboard more flexible and easier to explore.

STEP 6: Creating Bar Chart with Filters (Payment Value by Payment Type)

1. Open a **new worksheet (Sheet 4)** in Tableau.
2. Drag the **Payment Type** field from the Data Pane to the **Columns** shelf.
3. Drag the **Payment Value** measure to the **Rows** shelf.
4. Tableau will automatically create a **Bar Chart**.
5. Rename the worksheet title to "**Payment Value by Payment Type**".
6. Drag the **Payment Type** field to the **Filters** shelf.
7. In the **Filter [Payment Type]** dialog box, keep the required payment types selected (**boleto, credit_card, debit_card, voucher**) and click **OK**.
8. Right-click the **Payment Type** filter and select **Show Filter** to display it on the worksheet.
9. Drag the **Payment Value** field to the **Filters** shelf.
10. In the **Filter Field [Payment Value]** dialog box, select **At least** and click **Next**.
11. Enter the minimum payment value as **50** and click **OK**.

12. Right-click the **Payment Value** filter and select **Show Filter**.
13. Drag the **Order Purchase Timestamp** field to the **Filters** shelf.
14. In the **Filter Field [Order Purchase Timestamp]** dialog box, select **Years** and click **Next**.
15. Keep all available years (**2016, 2017, and 2018**) selected and click **OK**.
16. Right-click the **Year (Order Purchase Timestamp)** filter and select **Show Filter**.
17. Verify that the worksheet displays the **Payment Type**, **Payment Value**, and **Year** filters on the right side.
18. Confirm that the **Payment Value by Payment Type** bar chart updates interactively when any filter value is changed.



Filter [Payment Type]

General Wildcard Condition Top

☒ Select from list ☐ Custom value list ☐ Use all

Enter search text

- ☒ Null
- ☒ boleto
- ☒ credit_card
- ☒ debit_card
- ☐ not_defined
- ☒ voucher

All None ☐ Exclude

Summary

Field: [Payment Type]
Selection: Selected 5 of 6 values
Wildcard: All
Condition: None
Limit: None

Reset Apply Cancel OK

Filter [Payment Value]

Range of values At least At most Special

At least

50 1,25,42,084.19

50 1,25,42,084.19

Show: Only Relevant Values ☐ Include Null Values

Reset Apply Cancel OK

Filter Field [Order Purchase Timestamp]

How do you want to filter on [Order Purchase Timestamp]?

- ☐ Relative Date
- ☐ **Range of Dates**
- ☒ **Years**
- ☐ Quarters
- ☐ Months
- ☐ Days
- ☐ Hours
- ☐ Minutes
- ☐ Seconds
- ☐ Week numbers
- ☐ Weekdays
- ☐ Month / Year
- ☐ Month / Day / Year
- ☐ Individual Dates and Times

- ☐ Count
- ☐ Count (Distinct)
- ☐ Minimum
- ☐ Maximum

- ☐ Attribute

Filter [Year of Order Purchase Timestamp]

General Condition Top

☒ Select from list ☐ Custom value list ☐ Use all

Enter search text

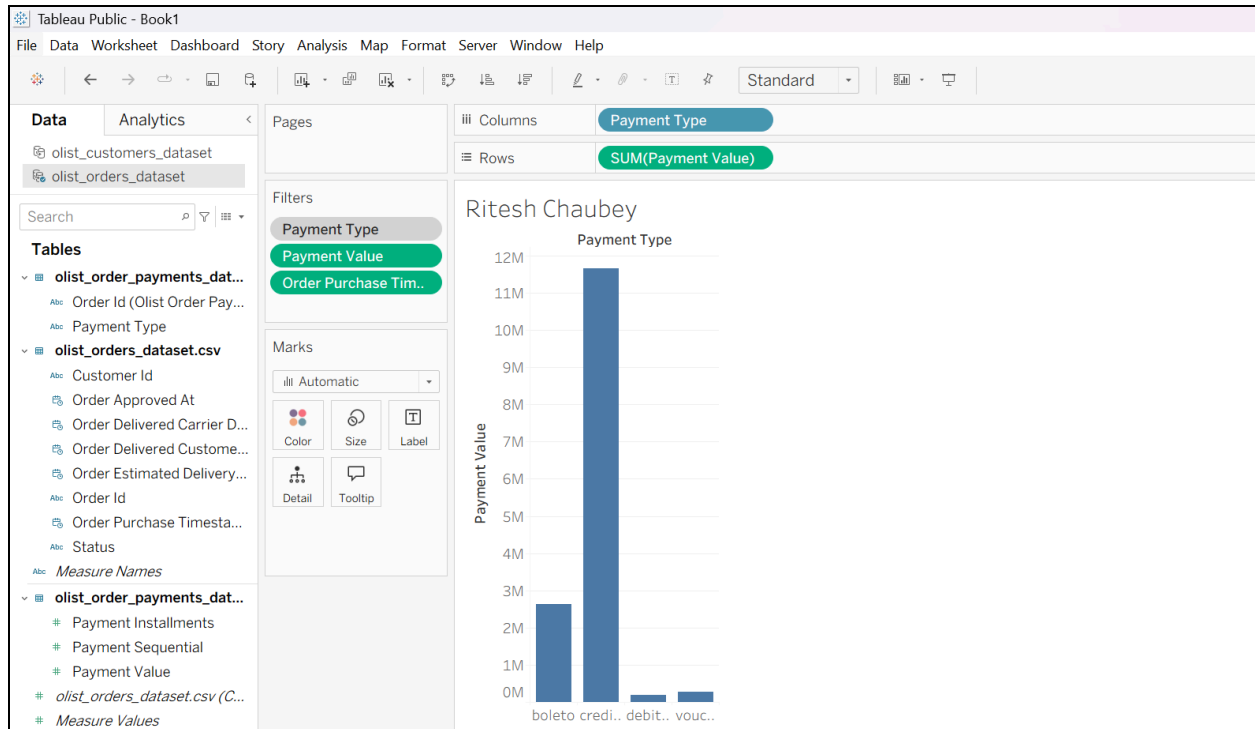
- ☒ 2016
- ☒ 2017
- ☒ 2018

☐ Exclude

Summary

Field: [Year of Order Purchase Timestamp]
Selection: Selected 3 of 3 values
Wildcard: All
Condition: None
Limit: None

☐ Filter to latest date value when workbook is opened



INSIGHT:

- Credit cards have the highest payment value among all payment types.
- Debit cards have the lowest payment value.
- Filters help analyze payment values by payment type and year.
- The chart updates instantly when filter values are changed.
- Interactive filters make data analysis easier and more flexible.

Step 7: Optimizing Dashboard Performance

1. Click the **Data Source** tab to return to the Data Source page.
2. Ensure that the **Extract** connection mode is selected.
3. Click the **Edit** option under the Extract section.
4. In the **Extract Filters** dialog box, click **Add Filter**.
5. Select the **Status** field and click **OK**.
6. In the **Filter [Status]** dialog box, select only **delivered** and uncheck all other status values.
7. Click **OK**, then click **Save Settings** to apply the extract filter.
8. Return to **Sheet 4**.
9. Right-click the **Payment Type** filter in the **Filters** shelf.
10. Select **Add to Context**.
11. Verify that the **Payment Type** filter changes to a **gray** color, indicating that it has been added as a **Context Filter**.
12. Confirm that the worksheet performance is optimized by using both the **Extract Filter** and the **Context Filter**.

Connections

olist_orders_dataset

Files

Use Data Interpreter

Data Interpreter might be able to clean your Text file workbook.

olist_customers_dataset.csv

olist_geolocation_dataset.csv

olist_order_items_dataset.csv

olist_order_payments_dataset.csv

olist_order_reviews_dataset.csv

olist_orders_dataset.csv

olist_products_dataset.csv

olist_sellers_dataset.csv

product_category_name_translation.csv

New Union

New Table Extension

olist_orders_dataset

Connection

Live

Extract

Edit

Refresh

Filters

0

Extract will contain all data.

olist_orders_datas...

Relate tables

Drag tables near olist_orders_dataset.csv to relate them

Add a base table

Drag out additional base tables for multi-fact analysis.

Need help adding data? [Learn more](#)

olist_orders_dataset.csv

14 fields 103887 rows

100 rows

Name olist_orders_dataset.csv

Description

No description available.

Fields

Type	Field Name	Phys...	Rem...
Abc	Order Id	olist_o...	order_id
Abc	Customer Id	olist_o...	custo...
Abc	Status	olist_o...	order_...
Abc	Order Purchase Timestamp	olist_o...	order_...
Abc	Order Approved At	olist_o...	order_...
Abc	Order Delivered Service Date	olist_o...	order_...

Abc	Abc	Abc	Abc
olist_orders_dataset.csv+	olist_orders_dataset.csv+	olist_orders_dataset.csv+	olist_orders_dataset.csv+
Order Id	Customer Id	Status	Order Purchase Time
b81ef226f3fe1789b1e8b2acac...	0a8556ac6be836b46b3e899...	delivered	25/04/2018 22:01:45
a9810da82917af2d9aefd1278...	f2c7fc58a9de810828715166c...	delivered	26/06/2018 11:01:38
25e8ea4e93396b6fa0d3dd7...	25b14b69de0b6e184ae6fe27...	delivered	12/12/2017 11:19:55
ba78997921bbcdc1373bb41e...	7a5d8efaaa1081f900628c30...	delivered	06/12/2017 12:04:06
42fdff880ba16b47b59251dd4...	15fd6fb8f8312dbb4674e4518...	delivered	21/05/2018 13:59:17
298fcdff173eb413e4d26d01b...	a24e6f72471e9dbafcb292bc3...	delivered	07/05/2018 13:20:41
771ee386b001f06208a7419e...	206c0fcc717333d169f95933...	delivered	23/06/2017 13:11:27
3d7239c394a212faae122962...	4d7a9b8bba459dce09d1b8fa...	delivered	05/06/2017 10:45:54
1f78449c97af54bf9e9fe88ba...	68c9daff1f93586c790b19c71...	delivered	22/07/2018 14:07:35

Data Source

Sheet 1

Sheet 2

Sheet 3

Sheet 4

Extract Data

Physical Tables

Store data using one table for each physical table.

Filters

No filters applied

Add Filter

Sta

Order Purchase Timestamp

Payment Installments

Status

Number of Rows

All Rows:

Top:

olist_orders_dat...

by

rows

Collapse All

Need help with extract settings? [Learn more](#)

History...

Hide All Unused Fields

Cancel

Save Settings

Filter [Status]

General

Wildcard

Condition

Top

Select from list

Custom value list

Use all

Enter search text

approved

canceled

created

☒ delivered

invoiced

processing

shipped

unavailable

All

None

Exclude

Summary

Field: [Status]

Selection: Selected 1 of 8 values

Wildcard: All

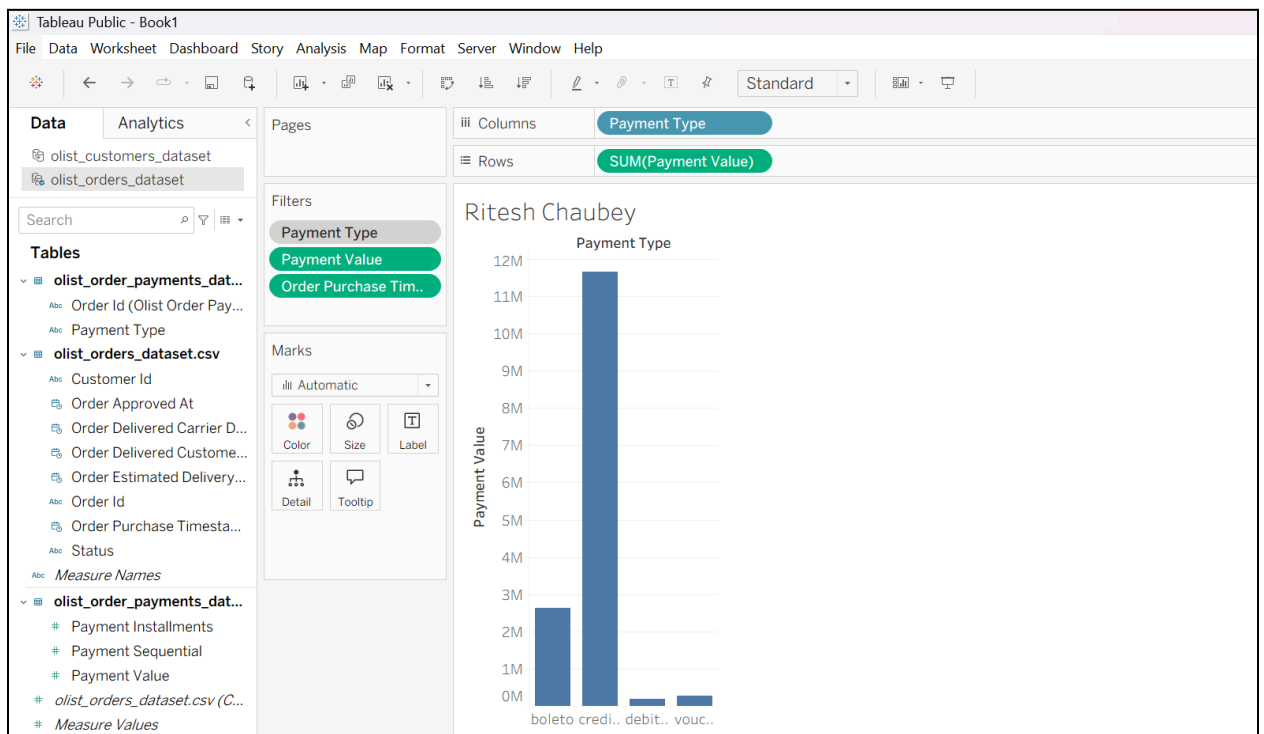
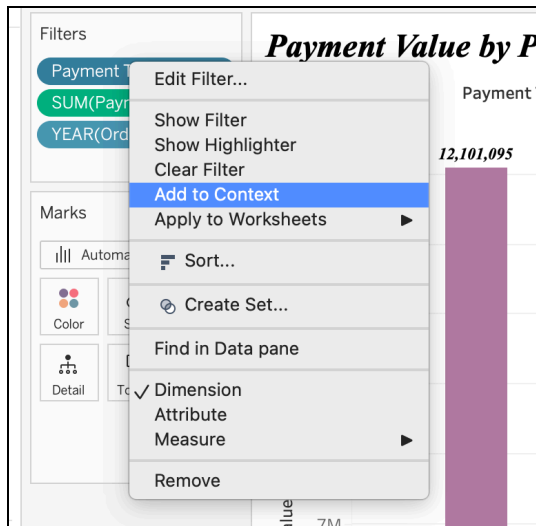
Condition: None

Limit: None

Reset

Cancel

OK



INSIGHT:

- **Credit Card** is the most frequently used payment method and contributes the highest total payment value.
- **Boleto, Debit Card, and Voucher** have comparatively lower payment values than Credit Card.
- Applying the **Status = Delivered** extract filter ensures that only successfully completed orders are included in the analysis.
- The **Payment Type, Payment Value, and Year** filters allow users to interactively analyze payment trends for different categories and time periods.

- Using a **Context Filter** improves dashboard performance by reducing the amount of data processed before applying other filters.

PRACTICAL INSIGHT:

- Different data sources such as **Orders, Payments, and Customers** can be connected and prepared efficiently in Tableau for analysis.
- **Metadata management, joins, and data blending** help combine related datasets and improve the quality and completeness of data analysis.
- **Dimension, measure, and date filters** enable users to explore the data interactively and focus on specific records based on their requirements.
- The **Payment Value by Payment Type** bar chart provides a clear comparison of payment methods, making it easier to identify the most significant payment trends.
- Using **Extract Filters** and **Context Filters** improves dashboard performance by reducing the amount of data processed while maintaining fast and efficient analysis.